

DEBT AND BANKRUPTCY

Jason Roderick Donaldson
USC & CEPR

Giorgia Piacentino
USC & CEPR & NBER

DONALDSON–GROMB–PIACENTINO 18

FACTS

Most ($> 90\%$) financing via debt

Debt heterogeneous: Collateral, covenants,...

Rights in bankruptcy; e.g. secured debt has priority

Rights outside bankruptcy; e.g. accelerate if covenant violated

DEBT VS. DEBT (NOT JUST DEBT VS. EQUITY)

Donaldson–Gromb–Piacentino I: Secured vs. unsecured in bankruptcy

DGP II: Secured vs. covenant-protected in and out of bankruptcy

DGP III: Secured vs. unsecured debt and asset buyers outside bankruptcy

Donaldson–Piacentino–Yu: Secured vs. unsecured in interbank networks

Donaldson–Morrison–Piacentino–Yu: Dispersed unsecured vs. one another

Donaldson–Koont–Piacentino–Vanasco: Loans vs. credit lines

SLIDE ON LESSONS

Different roles of collateral

One-creditor/exclusive models might not be enough

Allowing for multiple creditors can reverse results in literature

Collateral can backfire making unsecured borrowing harder

Covenants and collateral complements

More collateral can undermine asset allocation

Interbank networks can be robust but never fragile

Credit lines not liquidity insurance

FACTS

Collateral matters

Current theories suggest collateral matters for low pledgeability

“Collateral pledging makes up for a lack of pledgeable cash”

E.g. weak legal system, low creditor rights, low reputation

But collateral also matters when pledgeability is high

Interbank markets, syndicated loans, etc.

E.g. strong law, creditor rights, regulation, reputation

QUESTIONS

Why does collateral matter when pledgeability is high?

And is collateral always good for borrowers?

ROLE OF COLLATERAL

Role of collateral in most finance papers

Mitigate enforcement problem between borrower and creditor

Role of collateral in this paper

Mitigate enforcement problem among creditors

These roles correspond to two components of property rights

“Right of access”: right to seize collateral

“Right of exclusion”: right to stop others seizing collateral

THIS PAPER

Model of sequential financing based on three key assumptions

Assumption 1: Pledgeability is limited

Can divert a fraction of cash flows

Assumption 2: Contracts are non-exclusive

Can't commit not to borrow from third party

Assumption 3: Assets can be collateralized

Collateralized assets cannot be pledged to third party

LAWYERS' VIEW

“A secured transaction is the protection...against the claims of competing creditors”

—Kronman and Jackson (1979)

“Borrowers...may protect lenders against dilution by issuing secured debt”

—Schwarz (1997)

RESULTS

Paradox of pledgeability

Cannot borrow unsecured when pledgeability is high

Collateral rat race

Creditors require collateral to protect against collateral

Collateral overhang

Collateral prevents investment in positive NPV projects

MODEL

MODEL OVERVIEW

Three dates $t \in \{0, 1, 2\}$ and two states $s \in \{L, H\}$

s realized at Date 1, $\mathbb{P}[s = H] =: p$

Two riskless projects

Project 0 at Date 0

Project 1 at Date 1

At Date t , B can borrow from creditor C_t to invest in Project t

B can borrow secured (i.e. “collateralized”) or unsecured

PROJECTS

Project 0

Costs I_0 at Date 0

Pays off X_0 at Date 2

Project 1

Costs I_1^s at Date 1 in state s

Pays off X_1^s at Date 2

PLEDGEABILITY

Fraction θ of payoff is pledgeable

B can divert proportion $1 - \theta$ of project payoff

Creditors get up to θ of payoff according to priority

BORROWING AND INVESTMENT

B borrows from creditor C_t at Date t secured or unsecured

Secured debt

B can secure pledgeable payoff to a creditor

If B secures fraction σ , creditor gets exclusive claim to $\sigma\theta X$

Unsecured debt

B can promise pledgeable payoff unsecured

But B may collateralize projects to another creditor

CONTRACTING ENVIRONMENT

1. Courts treat secured debt as senior

“the absolute priority rule describes the basic order of payment in bankruptcy. Secured creditors get paid first, unsecured creditors get paid next”

—Lubben (2016)

2. B cannot commit not to collateralize

“the secured party whose presence violates the [negative pledge] covenant is entitled to repayment from the collateral before the injured negative pledgee”

—Bjerre (1999)

3. Collateral is not state contingent

C_0 is there at Date 0, but not at Date 1

TIMELINE

Date 0 B borrows I_0 from C_0 secured or unsecured

 If borrows, B invests in Project 0

Date 1 State s is revealed

 B borrows I_1^s from C_1 secured or unsecured

 If borrows, B invests in Project 1

Date 2 Projects payoff, repayments made, players consume

PARAMETER RESTRICTIONS

PARAMETER RESTRICTIONS

1. Pledgeable fraction of Project 0 is large enough to repay I_0

$$(1 - p)\theta X_0 > I_0$$

2. Project 1 has positive NPV in $s = H$ and negative NPV in $s = L$

$$X_1^H > I_1^H \quad \text{and} \quad X_1^L < I_1^L$$

3. Combined pledgeable cash flow less than costs in both states

$$\theta(X_0 + X_1^s) \leq I_0 + I_1^s$$

4. But greater than cost of Project 1 in state H

$$\theta(X_0 + X_1^H) \geq I_1^H$$

RESULTS

BENCHMARK: FIRST BEST

BENCHMARK: FIRST BEST

Project undertaken iff positive NPV

Date 0: Invest in Project 0

Date 1, state H : Invest in Project 1

Date 1, state L : Do not invest in Project 1

OVER-INVESTMENT PROBLEM

OVER-INVESTMENT PROBLEM

B always wants to invest in Project 1

Suppose B borrows secured from C_1

Dilutes any unsecured debt B has to C_0

B transfers cost of Project 1 to C_0

B thus captures PV of Project 1, not NPV

B borrows and invests even if negative NPV

RESULT 1: UNSECURED DEBT ACHIEVES FB FOR LOW θ

UNSECURED DEBT ACHIEVES FB FOR LOW θ

B always wants to invest at Date 1 so FB attained unsecured iff

Unconstrained in state H :

$$\theta(X_0 + X_1^H) \geq I_1^H$$

But constrained in state L :

$$\theta(X_0 + X_1^L) < I_1^L$$

B always unconstrained in H ; B constrained in L iff

$$\theta < \theta^* := \frac{I_L}{X_0 + X_1^L}$$

FB attained with unsecured debt iff pledgeability low ($\theta < \theta^*$)

RESULT 2: PARADOX OF PLEDGEABILITY

PARADOX OF PLEDGEABILITY

Increasing pledgeability relaxes borrowing constraint with C_1

Standard effect of pledgeability

Increasing pledgeability tightens borrowing constraint with C_0

New effect of pledgeability

PARADOX OF PLEDGEABILITY

Suppose θ is high

If C_0 lends unsecured, B dilutes C_0 in $s \in \{L, H\}$

C_0 is not repaid in either state

So C_0 will not lend unsecured for high pledgeability

RESULT 3: COLLATERAL RAT RACE

COLLATERAL RAT RACE

C_0 requires collateral as protection against dilution

Collateralization protects against collateralization

COLLATERAL RAT RACE

If B collateralizes σ_0 of Project 0, FB attained iff

Unconstrained in state H :

$$\theta \left((1 - \sigma_0) X_0 + X_1^H \right) \geq I_1^H$$

But constrained in state L :

$$\theta \left((1 - \sigma_0) X_0 + X_1^L \right) < I_1^L$$

or

$$\frac{I_1^H - \theta X_1^H}{\theta X_0} \leq 1 - \sigma_0 < \frac{I_1^L - \theta X_1^L}{\theta X_0}$$

Feasible for some $\sigma_0 \in [0, 1]$ whenever I_1^H not too large

RESULT 4: COLLATERAL OVERHANG

COLLATERAL OVERHANG

If I_1^H is large, can't attain first best

B constrained in state H

Collateralization prevents borrowing and efficient investment

Pledgeability causes “asset encumbrance”—collateral overhang

“Asset encumbrance not only poses risks to unsecured creditors...but also has wider...implications since encumbered assets are generally not available to obtain...liquidity”

PLEDGEABILITY VS. COLLATERALIZABILITY

PLEDGEABILITY VS. COLLATERALIZABILITY

Suppose fraction of a project is pledgeable but not collateralizable

Can be seized in the future

but hard to assign property rights to today

E.g. assets built while doing project, don't even exist at inception

Specifically B can collateralize at most μ_t of Project t at Date t

I.e. $\sigma_t \leq \mu_t$, so B collateralizes at most $\mu_t \theta X_t$

RESULT 5: COLLATERAL DAMAGE

COLLATERAL DAMAGE

First best is attained only if μ_1 is sufficiently small

High μ_1 makes it easier to borrow collateralized at Date 1

Triggers collateral rat race

Higher μ_1 means μ_0 must be higher to protect against dilution

More collateral used at Date 1, more required at Date 0

Collateral demand may be increasing in collateral supply

TWO ROLES OF COLLATERAL

TWO ROLES OF COLLATERAL

Reliance on collateral is u-shaped in θ

Low θ : classical role of collateral dominates

Collateralize to make up for lack of pledgeable cash

High θ : new role of collateral dominates

Collateralize to protect against dilution

CONCLUSIONS

CONCLUSIONS

Collateral protects creditors against the claims of other creditors

Paradox of pledgeability

- High pledgeability makes it easier to dilute

- Induces collateral rat race

Can't do projects due to collateral overhang—asset encumbrance

More collateral may decrease efficiency—collateral damage

THE PARADOX OF PLEDGEABILITY

DONALDSON–GROMB–PIACENTINO I: CODA

CRITIQUE

Paper extending analysis of our baseline model

- (i) Shows increasing θ cannot decrease welfare in our baseline model
- (ii) Suggests statements about collateral backfiring unfounded

But (i) consistent with our results and (ii) true in extension (μ_1)

We showed “collateralizability” (μ_1) can hurt in extension

They showed “pledgeability” (θ) cannot in baseline

CAN PLEDGEABILITY HURT IN EXTENSION?

Let $p = 0$ and as when B can commit not to invest at Date 1:

$$\mu_1 \theta X_1^L + \frac{1}{2} \theta ((1 - \mu_0) X_0 + (1 - \mu_1) X_1^L) < I_1^L \text{ iff } \theta < \theta^{**}$$

Overall: Collateral can hurt via θ or μ

Risk of dilution in future can undermine debt capacity today

DONALDSON–GROMB–PIACENTINO II (PRIORITIES)

FACTS

Firms rely on different types of debt at once

Including secured and unsecured with and without covenants

Secured debt has “absolute priority”

Unsecured debt has covenants limiting new secured debt

Negative pledge covenants common (e.g. 44% in Billett et al 07)

If covenants violated, right to accelerate debt

LAWYERS' VIEW ON N.P. COVENANTS

Negative pledge covenants may be of little practical comfort

The secured party whose presence violates the covenant is entitled to repayment from the collateral before the injured negative pledgee

The covenant does not prevent third parties from acquiring a security interest, but [is] merely...a hollow promise, for in the very act of breaching the covenant, the borrower places its assets out of reach of the negative pledgee

—Bjerre (1999)

LAWYERS' VIEW ON N.P. COVENANTS

Negative pledge covenants may be of little practical comfort

The secured party whose presence violates the covenant is entitled to repayment from the collateral before the injured negative pledgee

The covenant does not prevent third parties from acquiring a security interest, but [is] merely...a hollow promise, for in the very act of breaching the covenant, the borrower places its assets out of reach of the negative pledgee

—Bjerre (1999)

In the case of a debtor...indebted to secured creditors acceleration by unsecured creditors...seems somewhat futile

—Hahn (2010)

QUESTIONS

Q1. Why rely so heavily on negative pledge covenants?

Why not just use secured debt to promise priority credibly?

Q2. Why do borrowers use a multi-tiered debt structure?

Why mix secured/unsecured debt with/without covenants?

Q3. Why do some claims have absolute priority in the first place?

Even though it undermines other contracts?

THIS PAPER

Model of sequential financing based on two frictions

1. Limited pledgeability: can't borrow against projects' full PV
2. Contracts are non-exclusive: can sign conflicting contracts

Role of collateral: establish priority among conflicting contracts

Secured debt has absolute priority over collateral

New secured debt dilutes existing unsecured debt

DILUTION HAS TWO SIDES

Dilution can be bad—can lead to over-investment

New investments subsidized at expense of existing creditors

Dilution can be good—can prevent under-investment

Loosens borrowing constraints due to limited pledgeability

DILUTION HAS TWO SIDES

Dilution can be bad—can lead to over-investment

New investments subsidized at expense of existing creditors

Dilution can be good—can prevent under-investment

Loosens borrowing constraints due to limited pledgeability

Optimal debt structure allows good dilution, blocks bad dilution

MAIN RESULTS

- . Unsecured debt can't block dilution—over-investment
- . Secured debt blocks dilution, but maybe too much—under-investment
- . N.p. covenants allow dilution, but acceleration threat deters some

Acceleration futile to reverse dilution by new secured debt

But embeds option to dilute other unsecured debt

MAIN RESULTS

- . Unsecured debt can't block dilution—over-investment
- . Secured debt blocks dilution, but maybe too much—under-investment
- . N.p. covenants allow dilution, but acceleration threat deters some

Acceleration futile to reverse dilution by new secured debt

But embeds option to dilute other unsecured debt

Right debt structure implements efficiency under current priority rule

RESULTS MATTER FOR POLICY

Lawyers advocate relaxing absolute priority of secured debt

This article challenges the desirability of...full priority of secured claims
—Bebchuk–Fried (1996)

Such proposals protect against dilution, but maybe too much

We show existing priority rules efficient if right mix of debt

RESULTS RESONATE WITH PRACTICE

Explain why negative pledge covenants pervasive

Billet et al 07, Ivashina–Vallée 18

Explain why covenants frequently violated

Chava–Roberts 08, Dichev–Skinner 02, Roberts–Sufi 09

Explain why covenants typically waived following violations

Beneish–Press 93/95, Gopalakrishnan–Parkash 95, Sweeney 94

Explain why covenant use increases in growth opportunities

Billet–King–Mauer 07

Explain why distressed firms use secured debt

Badoer et al. 17, Barclay–Smith 95, Rauh–Sufi 10

LITERATURE

Models of collateral establishing priority Ayotte–Bolton 11, DeMarzo 19, Donaldson–Gromb–Piacentino 19, Stulz–Johnson 85

Focus on priority in bankruptcy, here also outside bankruptcy

Models of covenants Attar et al. 14, Ayotte–Bolton 11, Garleânu– Zwiebel 09, Park 00, Rajan–Winton 95

Focus on debt vs. equity, here debt vs. debt

MODEL

MODEL OVERVIEW

Three dates: $t \in \{0, 1, 2\}$

Risk-neutral players: penniless borrower B; competitive creditors

Projects: Project 0 available at Date 0, Project 1 at Date 1

At Date 1, quality of Project 1 revealed $Q \in \{H, L\}$, $\mathbb{P}[H] =: q$

At Date 2, projects succeed or fail, $\mathbb{P}[\text{success}] =: p$ (corr. = 1)

At Date t , B can borrow from creditors to fund Project t

PROJECTS

Project 0

Costs I_0 at Date 0

Pays off $X_0 + Y_0$ at Date 2 if succeeds, zero if fails

X_0 is pledgeable, Y_0 not

Project 1

Costs I_1 at Date 1

Pays off $X_1^Q + Y_1^Q$ at Date 2 if succeeds, zero if fails

X_1^Q is pledgeable, Y_1^Q not

Can liquidate at Date 1 for expected pledgeable c.f. pX

INSTRUMENTS

Secured debt: promise to repay F^{sec} secured by projects as collateral

Unsecured debt: promise to repay without collateral

Unsecured debt with n.p. covenants: promise without collateral

But option to accelerate if B borrows secured

PRIORITY RULE

Secured debt has priority over collateral

Ahead of all unsecured debt (absolute priority rule)

Ahead of later secured debt (first-in-time rule)

Ahead of any other claimants if collateral liquidated/sold

Unsecured pro-rata in default

Acceleration gives effective priority over other unsec. debt

But not over secured debt (protected by collateral)

PRIORITY RULE

Secured debt has priority over collateral

Ahead of all unsecured debt (absolute priority rule)

Ahead of later secured debt (first-in-time rule)

Ahead of any other claimants if collateral liquidated/sold

Unsecured pro-rata in default

Acceleration gives effective priority over other unsec. debt

But not over secured debt (protected by collateral)

Modeled as sequential service constraint on unsecured debt

PRIORITY RULE REFLECTS PRACTICE

Current law...rests on three “priority principles”: First, if the first creditor...makes an unsecured loan, it shares pro rata with later unsecured creditors...on default. Second, if this initial creditor makes an unsecured loan and a later creditor takes security, the later creditor has priority over the initial creditor.... Third, if the initial creditor makes a secured loan, it generally has priority over later creditors

—Schwartz (1989)

PRIORITY RULE REFLECTS PRACTICE

Current law...rests on three “priority principles”: First, if the first creditor...makes an unsecured loan, it shares pro rata with later unsecured creditors...on default. Second, if this initial creditor makes an unsecured loan and a later creditor takes security, the later creditor has priority over the initial creditor.... Third, if the initial creditor makes a secured loan, it generally has priority over later creditors

—Schwartz (1989)

The security interest gives the lender...a “priority right”...in the face of competing claims of purchasers, transferees, and other creditors

—Merrill-Smith (2001)

PRIORITY RULE REFLECTS PRACTICE

Current law...rests on three “priority principles”: First, if the first creditor...makes an unsecured loan, it shares pro rata with later unsecured creditors...on default. Second, if this initial creditor makes an unsecured loan and a later creditor takes security, the later creditor has priority over the initial creditor.... Third, if the initial creditor makes a secured loan, it generally has priority over later creditors

—Schwartz (1989)

The security interest gives the lender...a “priority right”...in the face of competing claims of purchasers, transferees, and other creditors

—Merrill-Smith (2001)

[Acceleration] facilitates collection by the speedy creditors with the potential of harming the less fortunate ones.... But in the case of a debtor who is also indebted to secured creditors acceleration by unsecured creditors...seems somewhat futile

—Hahn (2010)

TIMELINE

Date 0 B funds Project 0 from competitive creditors or not

Date 1 Project 1's quality Q is revealed

B funds Project 1 from competitive creditors or not

Acceleration (hence liquidation) or continuation

Date 2 Payoffs realized and repayments made

PARAMETER RESTRICTIONS

R 1. Project 0 efficient; Project 1 efficient if high quality, not if low

$$p(X_0 + Y_0) > I_0$$

$$p(X_1^H + Y_1^H) > I_1 > p(X_1^L + Y_1^L)$$

R 2. Expected pledgeable c.f. of positive-NPV projects exceeds costs

$$pX_0 + qpX_1^H \geq I_0 + qI_1$$

R 3. Project 1 is not self-financing

$$pX_1^Q < I_1$$

R 4. Total liquidation value not too small

$$p(X_0 + X_1^Q) \geq \frac{I_1}{p}$$

PARAMETER RESTRICTIONS

R 1. Project 0 efficient; Project 1 efficient if high quality, not if low

$$p(X_0 + Y_0) > I_0$$

$$p(X_1^H + Y_1^H) > I_1 > p(X_1^L + Y_1^L)$$

R 2. Expected pledgeable c.f. of positive-NPV projects exceeds costs

$$pX_0 + qpX_1^H \geq I_0 + qI_1$$

R 3. Project 1 is not self-financing

$$pX_1^Q < I_1$$

R 4. Total liquidation value not too small

$$p(X_0 + X_1^Q) \geq \frac{I_1}{p} = F_1^{\text{sec}}$$

FIRST BEST

FIRST BEST

Project undertaken iff positive NPV

Date 0: invest in Project 0

Date 1: invest in Project 1 if high quality, not if low

TWO THINGS NECESSARY FOR FIRST BEST

1. B does not invest in Project 1 if $Q = L$

Problem: non-exclusivity can lead to over-investment

Borrowing constraints too loose: may need to block dilution

2. B can invest in Project 1 if $Q = H$

Problem: limited pledgeability can lead to under-investment

Borrowing constraints too tight: may need dilution

BENCHMARK: EXCLUSIVE CONTRACTS

BENCHMARK: EXCLUSIVE CONTRACTS

Suppose limited pledgeability

B may not be able to borrow against projects' full PV

But suppose exclusivity

B can't borrow from third party

Can write bilateral contingent contract

RESULT 0: EXCLUSIVE CONTRACT IMPLEMENTS FB

EXCLUSIVE CONTRACT IMPLEMENTS FB

Two things necessary for FB

1. B does not invest in Project 1 if $Q = L$

With exclusivity, can't dilute with new debt to third party

No incentive to over-invest

2. B invests in Project 1 if $Q = H$ (overcoming non-pledgeab.)

By PR 3, Project 1 is not self-financing: $pX_1^H < I_1$

By PR 2, creditor still breaks even on average

Creditor takes loss if $Q = H$ but makes offsetting gain if L

NON-EXCLUSIVE CONTRACTS

UNSECURED DEBT

RESULT 1: INEFFICIENCY OF UNSECURED DEBT

RESULT 1: INEFFICIENCY OF UNSECURED DEBT

Say B takes on unsecured debt F_0^u at Date 0

Unsecured debt can be diluted: over-investment if $Q = L$

Too much bad dilution

SECURED DEBT

SECURED DEBT CAN BLOCK DILUTION

Say B takes on secured debt F_0^{sec} at Date 0 (rest unsecured)

Secured debt F_0^{sec} cannot be diluted, unsecured can be

SECURED DEBT CAN BLOCK DILUTION

Say B takes on secured debt F_0^{sec} at Date 0 (rest unsecured)

Secured debt F_0^{sec} cannot be diluted, unsecured can be

What is B's debt capacity at Date 1?

B'S DEBT CAPACITY

B's debt capacity is max B can borrow (secured)

Can promise new sec. creditors everything not promised as F_0^{sec}

New secured debt paid ahead of Date-0 unsec. debt (APR)

But paid after Date-0 secured debt (first-in-time rule)

$$\text{Debt capacity} = p\left(X_0 + X_1^Q - F_0^{\text{sec}}\right)$$

IS FB IMPLEMENTABLE WITH SECURED DEBT?

FB implemented if constrained if $Q = L$ but not if $Q = H$:

1. Constrained if $Q = L$:

$$\underbrace{p(X_0 + X_1^L - F_0^{\text{sec}})}_{\text{debt capacity}} < I_1$$

2. But unconstrained if $Q = H$:

$$\underbrace{p(X_0 + X_1^H - F_0^{\text{sec}})}_{\text{debt capacity}} \geq I_1$$

or

$$X_1^L < \frac{I_1 - p(X_0 - F_0^{\text{sec}})}{p} \leq X_1^H$$

RESULT 2: SECURED DEBT IMPLEMENTS FB IF X_1^H HIGH

SECURED DEBT IMPLEMENTS FB IF X_1^H HIGH

Financing Project 1 requires dilution; can't if all debt is secured

Reduce secured debt to allow dilution

But could allow for too much dilution

If $X_1^L < X_1^H$ can find F_0^s s.t. can fund H project, not L project

Because less dilution needed to fund H than L

SECURED DEBT IMPLEMENTS FB IF X_1^H HIGH

Financing Project 1 requires dilution; can't if all debt is secured

Reduce secured debt to allow dilution

But could allow for too much dilution

If $X_1^L < X_1^H$ can find F_0^s s.t. can fund H project, not L project

Because less dilution needed to fund H than L

What if need to dilute more to fund H than L project?

UNDER-INVESTMENT WITH SECURED DEBT

Suppose need more dilution to fund H project than L ($X_1^H < X_1^L$)

Any F_0^{sec} blocking funding L project blocks H project too

Secured borrowing blocks future positive-NPV projects

Secured borrowing “encumbers assets”

Asset encumbrance not only poses risks to unsecured creditors...but also has wider...implications since encumbered assets are generally not available to obtain...liquidity

UNDER-INVESTMENT WITH SECURED DEBT

Suppose need more dilution to fund H project than L ($X_1^H < X_1^L$)

Any F_0^{sec} blocking funding L project blocks H project too

Secured borrowing blocks future positive-NPV projects

Secured borrowing “encumbers assets”

Asset encumbrance not only poses risks to unsecured creditors...but also has wider...implications since encumbered assets are generally not available to obtain...liquidity

There is a collateral overhang

CAN RENEGOTIATION SOLVE PROBLEM?

High-quality project creates surplus $p(X_1^H + Y_1^H) > I_1$

Should find Coasian bargain to fund it

But limited pledgeability can make renegotiation infeasible

Renegotiation must make everyone (strictly) better off

But by PR 3, pledgeable c.f. is less than cost, $pX_1^H < I_1$

So can't credibly allocate enough to creditors

NEGATIVE PLEDGE COVENANTS

NEGATIVE PLEDGE COVENANTS

Say B takes on unsecured debt with covenants F_0^{cov} at Date 0

Say B violates covenant, funding Project 1 with secured debt F_1^{sec}

Assume F_1^{sec} large enough that can't repay F_0^{cov} (wlog)

Given violation, creditor with n.p. can accelerate, forcing liquidation

$$\min \left\{ F_0^{\text{cov}}, p(X_0 + X_1^Q) - F_1^{\text{sec}} \right\} = p(X_0 + X_1^Q) - F_1^{\text{sec}}$$

NEGATIVE PLEDGE COVENANTS

Say B takes on unsecured debt with covenants F_0^{cov} at Date 0

Say B violates covenant, funding Project 1 with secured debt F_1^{sec}

Assume F_1^{sec} large enough that can't repay F_0^{cov} (wlog)

Given violation, creditor with n.p. can accelerate, forcing liquidation

$$\underbrace{\min \left\{ F_0^{\text{cov}}, p(X_0 + X_1^Q) - F_1^{\text{sec}} \right\}}_{\text{acceleration value}} = p(X_0 + X_1^Q) - F_1^{\text{sec}}$$

NEGATIVE PLEDGE COVENANTS

Say B takes on unsecured debt with covenants F_0^{cov} at Date 0

Say B violates covenant, funding Project 1 with secured debt F_1^{sec}

Assume F_1^{sec} large enough that can't repay F_0^{cov} (wlog)

Given violation, creditor with n.p. can accelerate, forcing liquidation

$$\underbrace{\min \left\{ F_0^{\text{cov}}, \underbrace{p(X_0 + X_1^Q)}_{\text{liquidation value}} - F_1^{\text{sec}} \right\}}_{\text{acceleration value}} = p(X_0 + X_1^Q) - F_1^{\text{sec}}$$

ACCELERATION THREAT CAN DETER DILUTION

B won't dilute if anticipates exercise of acceleration option

Forces liquidation and destroys non-pledgeable c.f. Y

ACCELERATION THREAT CAN DETER DILUTION

B won't dilute if anticipates exercise of acceleration option

Forces liquidation and destroys non-pledgeable c.f. Y

But is acceleration threat credible?

IS ACCELERATION THREAT CREDIBLE?

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} \not\geq \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} \not\geq \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

Acceleration makes secured debt safer (paid first out of liquidation)

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} \not\geq \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

Acceleration makes secured debt safer (paid first out of liquidation)

Subsidy to secured debt, “tax” on accelerated debt

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} \not\geq \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

Acceleration makes secured debt safer (paid first out of liquidation)

Subsidy to secured debt, “tax” on accelerated debt

Don’t accelerate to avoid tax

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} \not\geq \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

Acceleration makes secured debt safer (paid first out of liquidation)

Subsidy to secured debt, “tax” on accelerated debt

Don’t accelerate to avoid tax

Covenants don’t discipline: same outcome as without covenants

IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} \not\geq \underbrace{p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

Acceleration makes secured debt safer (paid first out of liquidation)

Subsidy to secured debt, “tax” on accelerated debt

Don’t accelerate to avoid tax

Covenants don’t discipline: same outcome as without covenants

What if fraction ϕ of unsecured with n.p. covenants (rest unsec.)?

$\phi < 1$: IS ACCELERATION THREAT CREDIBLE?

$\phi < 1$: IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{\phi p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

$\phi < 1$: IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p\left(X_0 + X_1^Q\right) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{\phi p\left(X_0 + X_1^Q - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

$\phi < 1$: IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p(X_0 + X_1^Q) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{\phi p(X_0 + X_1^Q - F_1^{\text{sec}})}_{\text{continuation value}}$$

Acceleration doesn't undo harm of dilution with secured debt

But dilutes other unsecured debt $(1 - \phi)$

Threat can be credible if $(1 - \phi)$ large

$\phi < 1$: IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p(X_0 + X_1^Q) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{\phi p(X_0 + X_1^Q - F_1^{\text{sec}})}_{\text{continuation value}}$$

Acceleration doesn't undo harm of dilution with secured debt

But dilutes other unsecured debt $(1 - \phi)$

Threat can be credible if $(1 - \phi)$ large

Yet another side of dilution: to make acceleration credible

$\phi < 1$: IS ACCELERATION THREAT CREDIBLE?

Yes, if

$$\underbrace{p(X_0 + X_1^Q) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{\phi p(X_0 + X_1^Q - F_1^{\text{sec}})}_{\text{continuation value}}$$

Acceleration doesn't undo harm of dilution with secured debt

But dilutes other unsecured debt $(1 - \phi)$

Threat can be credible if $(1 - \phi)$ large

Yet another side of dilution: to make acceleration credible

If credible at the right time could lead to efficient investment

FB IMPLEMENTABLE WITH N.P. COVENANTS?

FB implemented if acceleration threat credible if L , not if H :

1. If $Q = L$, accelerate given violation

$$\underbrace{p\left(X_0 + X_1^L\right) - F_1^{\text{sec}}}_{\text{acceleration value}} > \underbrace{\phi p\left(X_0 + X_1^L - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

2. If $Q = H$, don't accelerate given violation

$$\underbrace{p\left(X_0 + X_1^H\right) - F_1^{\text{sec}}}_{\text{acceleration value}} < \underbrace{\phi p\left(X_0 + X_1^H - F_1^{\text{sec}}\right)}_{\text{continuation value}}$$

or

$$X_1^L > \frac{(1 - p\phi)}{p(1 - \phi)} F_1^{\text{sec}} - X_0 \geq X_1^H$$

RESULT 3: COVENANTS IMPLEMENT FB IF X_1^H SMALL

COVENANTS IMPLEMENT FB IF X_1^H SMALL

Complement of when secured debt implements FB

If need to dilute more to fund H than L , can find ϕ to implement FB

If $Q = L$, don't invest: covenant upheld, deterring dilution

If $Q = H$, invest: covenant waived, allowing dilution

COVENANTS IMPLEMENT FB IF X_1^H SMALL

Complement of when secured debt implements FB

If need to dilute more to fund H than L , can find ϕ to implement FB

If $Q = L$, don't invest: covenant upheld, deterring dilution

If $Q = H$, invest: covenant waived, allowing dilution

Acceleration less attractive for large dilution

DON'T ACCELERATE IF DILUTION LARGE

After covenants violated, n.p. debt is endogenously junior

Paid after new secured debt, ahead of equity

Both debt-like and equity-like

More equity-like if dilution is large (closer to residual claim)

Equity is a call option on the assets

Do not want to exercise (accelerate and liquidate) early

“Gamble for resurrection” like firm in distress

IS ACCELERATION RENEGOTIATION PROOF?

Liquidation following covenant violation destroys surplus

Can renegotiation reallocate this surplus to avoid liquidation?

No! Liquidation destroys only non-pledgeable c.f.

Can't allocate pledgeable c.f. to make everyone (strictly) better off

OPTIMAL DEBT STRUCTURE

OPTIMAL DEBT STRUCTURE

Can find debt structure that implements FB

Avoids under-investment but block over-investment

I.e. manages trade off between financial rigidity and flexibility

OPTIMAL DEBT STRUCTURE

Can find debt structure that implements FB

Avoids under-investment but block over-investment

I.e. manages trade off between financial rigidity and flexibility

Mixes secured and unsec. debt with and w/o cov.

DEBT STRUCTURE REFLECTS PRACTICE

DEBT STRUCTURE REFLECTS PRACTICE

1. Negative pledge covenants are common
Billett et al. 07, Ivashina-Vallée
2. Covenants are frequently violated
Chava-Roberts 08, Nini-Smith-Sufi 12, Roberts-Sufi 09
3. Following violations, covenants are typically waived
Gopalakrishnan-Prakash 95, Nini-Smith-Sufi 12
4. Multi-layered: with and without covenants secured and unsecured
Rauh-Sufi 10
5. No pecking order of debt types
Kermani-Ma 19, Rampini-Viswanathan 18, Rauh-Sufi 10
6. Secured debt reduces future investment (collateral overhang)
Badoer-Dudley-James 17
7. Public and private debt coexist; private debt has more covenants
Gopalakrishnan and Prakash 95

NEW TESTABLE IMPLICATIONS

TESTABLE IMPLICATIONS

1. Firms more exposed to under-investment should use covenants

Billett et al. 17: covenant use increases in growth opportunities

2. Firms more exposed to over-investment should use collateral

Badoer et al. 19: secured debt use increases in distress

3. Firms use secured debt less when good investments less pledgeable

Benmelech–Kumar–Rajan 19: more intangibles, less collateral

HOW ARE COVENANTS PRICED?

HOW ARE COVENANTS PRICED?

Empirically covenants are valuable

Matvos 13, Bradley–Roberts 14, Green 18

Here too: with covenants, do Project 1 if $Q = H$, deterred if $Q = L$

Still, unsecured debt with and without covenants priced equally

Acceleration is off equilibrium, but disciplines B

All unsecured creditors benefit from credible threat

Bradley–Roberts: covenants in firms' loans reduce yield on bonds

COVENANTS VS. COLLATERAL

COVENANTS VS. COLLATERAL

Literature stressed substitutability of covenants and collateral

Secured debt and covenants are substitutes (both are issued to protect against dilution)

—Schwartz (1989)

We show that there is also complementarity

Covenants implement FB only in conjunction with collateral

Need covenants to promise not to use collateral

But also need collateral to break that promise

SHORT-TERM DEBT VS. COVENANTS

SHORT-TERM DEBT VS. COVENANTS

Acceleration threat kind of discipline via short term debt

Two features new to literature on disciplining short-term debt

1. Mix types of debt so discipline state-contingent

Demand redemption of ST debt only to dilute LT debt

2. Maturity of covenant-protected debt depends on violation

If always short term, demand even if no violation/investment

$$\min \{ pX_0, F_0^{\text{ST}} \} \geq pF_0^{\text{ST}}$$

DEBT VS. DEBT

DEBT VS. DEBT

Literature stresses how covenants address debt-equity conflicts

In this paper, we examine how debt contracts are written to control the bondholder-stockholder conflict. We investigate...covenants...

—Smith–Warner (1979)

We argue debt-debt conflicts as important for some covenants

E.g. negative pledge covenants!

CREDITORS VS. CREDITORS

CREDITORS VS. CREDITORS

Need multiple Date-0 creditors

Pit creditors against each other to make acceleration credible

N.p. debt held by large, strategic creditor (e.g. bank)

Needs to be able to waive covenant

Unsecured debt held by passive creditors (e.g. bondholders)

Indeed, in practice, private debt has more covenants

Barclay-Smith 95, Bradley-Roberts 14

MANY SIDES OF DILUTION

MANY SIDES OF DILUTION

Dilution via collateral: secured debt dilutes unsecured debt

Can be bad: causes over-investment (non-exclusivity)

Can be good: prevents under-investment (non-pledgeability)

Dilution via maturity: unsecured debt dilutes other unsec. debt

Can be bad: inefficient liquidation (acceleration)

Can be good: deters inefficient dilution (acceleration threat)

Optimal debt structure: choose F_0^{sec} and ϕ to get good w/o bad

IMPLEMENTATION WITH “SIMPLE” CONTRACTS

IMPLEMENTATION WITH “SIMPLE” CONTRACTS

FB outcome and repayments are state contingent

Investment and financing greater in one state than other

Implement FB with realistic, non-contingent debt

Debt contracts not state contingent, but debt structure is

Contingent transfers via prioritized, conflicting debts

EXTENSIONS AND ROBUSTNESS

EXTENSIONS AND ROBUSTNESS

1. Continuum of qualities
2. “Coalitional” renegotiation
3. Secured debt with covenants and deductibles
4. Dilution via pari passu debt
5. Financing via subordinated debt
6. Bankruptcy due to acceleration

CONTINUUM OF QUALITIES

CONTINUUM OF QUALITIES

Say Project 1 can be one of a continuum of qualities Q

Secured debt: fund projects with pledgeable c.f. above cutoff

F_0^{sec} limits debt capacity, block projects if $X_1^Q < X_1^{\text{sec}}$

N.p. covenants: fund projects with pledgeable c.f. below cutoff

$\phi < 1$ incentivizes acceleration, deters projects if $X_1^Q > X_1^{\text{cov}}$

Can implement FB as long as X_1^Q is monotonic in Q

If increasing, use secured debt to block low-quality projects

If decreasing, use covenants to deter low-quality projects

COALITIONAL RENEGOTIATION

COALITIONAL RENEGOTIATION

So far, ruled out coalitional renegotiation: B bribes creditor to waive

N.p. creditor would accept second lien as bribe not to accelerate:

$$\underbrace{p(X_0 + X_1^L - F_1^s)}_{\text{second-lien value}} \geq \underbrace{p(X_0 + X_1^L) - F_1^s}_{\text{acceleration value}}$$

(Hurts other creditors: not valid multilateral renegotiation)

Ruling it out realistic: debt has “inter-creditor agreements”

Prohibit creditors from altering debts at expense of others

Prohibit creditors from changing relative priority of their debts

(Go back)

SECURED DEBT WITH COVENANTS AND DEDUCTIBLES

SEC. DEBT WITH COVENANTS & DEDUCTIBLES

So far, ruled out taking new debt to pay accelerated debt by ass.

Possibility realistic to assume too late to go back to market

Can rule out it out endogenously with addition to debt structure

Add secured debt with n.p. covenant at Date 0

But with deductible s.t. can take new debt up to $F_1^s = I_1/p$

Because debt secured, always accelerates if violate

Realistic: covenants in sec. debt common, but less than in unsec.

Deductibles common too

DILUTION VIA PARI PASSU DEBT

CAN IT IMPLEMENT FIRST BEST?

DILUTION VIA PARI PASSU DEBT: FIRST BEST?

Suppose B borrows unsecured with leverage covenant

If leverage covenant violated, always acceleration

Accelerated debt is always paid first (unlike with n.p. cov)

Dilution with unsecured debt doesn't implement FB

Doesn't create state-contingencies

CAN DILUTION WITH UNSEC. DEBT CIRCUMVENT N.P. COV?

DILUTION VIA PARI PASSU DEBT

So far, implicitly ruled out dilution with new unsecured debt

Still, dilutes existing debt, w/o violating n.p. covenants

But easy to rule out with leverage covenant

No bad dilution via unsec. debt (need sec. debt for good dilution)

FINANCING VIA SUBORDINATED DEBT

FINANCING VIA SUBORDINATED DEBT

Covenants also deter B from doing even self-financing projects

E.g. $Q = HH$ with $pX_1^{HH} > I_1$

B won't invest if violates covenants, triggering acceleration

But leverage covenants typically allow for new subordinated debt

Subordinated debt solves under-investment if $Q = HH$

Fund Project 1 without violating covenant

FINANCING VIA SUBORDINATED DEBT

Covenants also deter B from doing even self-financing projects

E.g. $Q = HH$ with $pX_1^{HH} > I_1$

B won't invest if violates covenants, triggering acceleration

But leverage covenants typically allow for new subordinated debt

Subordinated debt solves under-investment if $Q = HH$

Fund Project 1 without violating covenant

(Secured debt solves underinvestment if pledgeability low)

BANKRUPTCY DUE TO ACCELERATION

BANKRUPTCY DUE TO ACCELERATION

So far accelerating creditor could get money out before bankruptcy

But if acceleration causes bankruptcy, then he is pari passu

Can implement first best choosing F_0^c s.t. when covenant is violated

1. If $Q = H$: acceleration causes bankruptcy, so no acceleration

$$p(X_0 + X_1^H) - F_1^s < F_0^c$$

2. If $Q = L$: acceleration doesn't cause bankruptcy, so accelerate

$$p(X_0 + X_1^L) - F_1^s \geq F_0^c$$

Can find F_0^c if $X_1^H < X_1^L$ as in baseline model

QUESTIONS

QUESTIONS

Q1. Why use negative pledge covenants instead of secured debt?

Q2. Why do borrowers use a multi-tiered debt structure instead?

Q3. Why do some claims have absolute priority in the first place?

ANSWERS

Q1. Why use negative pledge covenants instead of secured debt?

A1. Secured debt can protect too much against dilution

Q2. Why do borrowers use a multi-tiered debt structure instead?

A2. Allows efficient dilution and prevents inefficient dilution

Q3. Why do some claims have absolute priority in the first place?

A3. Allows borrowers to combine instruments to overcome frictions

CONCLUSIONS

CONCLUSIONS

Covenants can be violated and contracts can conflict

Need priority rule to resolve conflicts

Lawyers argue current priority rule is perverse

But we show it helps implement efficiency via right debt structure

Debt structure is multi-tiered—rich and realistic

CONFLICTING PRIORITIES:
A THEORY OF COVENANTS AND COLLATERAL